Reflection Document: Date Display GUI 2

1. Main Functionalities & Implementation The application is a graphical user interface tool designed to display a selected base date and calculate a future or past date based on user-defined inputs (amount, direction, and time unit). Additionally, it allows users to dynamically switch between multiple date formats (e.g., US, European, Vietnamese). The implementation utilizes JavaFX for the interface and the java.time.LocalDate class for accurate date calculations.

2. Design Decisions & MVC Architecture The application was strictly designed using the Model-View-Controller (MVC) architectural pattern to ensure the separation of concerns and maintainability. The **Model** (DateModel) encapsulates the core business logic and state, using JavaFX ObjectProperty<LocalDate> to hold date values. The **View** (DateView.fxml) defines the graphical layout declaratively, completely separated from Java logic. The **Controller** (DateController) bridges the two, injecting FXML elements and handling user interactions without containing the underlying calculation logic.

3. Improvements from the Previous Version In the previous AWT/Swing version, the components were tightly coupled. The Controller manually handled specific button clicks using a complex ActionListener with multiple conditional statements, and then explicitly instructed the View to update. Error handling relied on catching runtime exceptions and showing intrusive popup dialogs. In this JavaFX iteration, these limitations were resolved through reactive programming. Event handling is streamlined using @FXML methods, and validation is handled smoothly within the UI without blocking popups.

4. JavaFX Properties and Binding JavaFX properties and bindings were utilized to eliminate boilerplate synchronization code. A StringBinding was created using Bindings.createStringBinding() to observe both the selectedDate property and the dateFormat ComboBox value. This ensures the displayed date label updates automatically whenever the base date or the format changes. Furthermore, the disableProperty() of the Apply button was bound to the input text field; it automatically disables the button and displays an error message if the user enters non-numeric or invalid data, ensuring robust real-time validation.